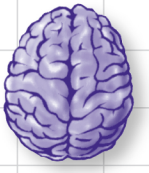
A Brain-Friendly Guide

Head First Python



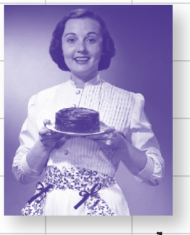
Load important programming concepts directly into your brain

Model data with lists, sets, and dictionaries

Preserve your data in a pickle

Hook up with JSON, Android, and App Engine

Move your custom app to the Web



Share your code with the world on PyPI

O'REILLY®

Paul Barry

Advance Praise for *Head First Python*

"*Head First Python* is a great introduction to not just the Python language, but Python as it's used in the real world. The book goes beyond the syntax to teach you how to create applications for Android phones, Google's App Engine, and more."

— David Griffiths, author and Agile coach

"Where other books start with theory and progress to examples, *Head First Python* jumps right in with code and explains the theory as you read along. This is a much more effective learning environment, because it engages the reader to *do* from the very beginning. It was also just a joy to read. It was fun without being flippant and informative without being condescending. The breadth of examples and explanation covered the majority of what you'll use in your job every day. I'll recommend this book to anyone starting out on Python."

— Jeremy Jones, coauthor of *Python for Unix and Linux System Administration*

"*Head First Python* is a terrific book for getting a grounding in a language that is increasing in relevance day by day."

— Phil Hartley, University of Advancing Technology

Praise for other *Head First* books

"Kathy and Bert's *Head First Java* transforms the printed page into the closest thing to a GUI you've ever seen. In a wry, hip manner, the authors make learning Java an engaging 'what're they gonna do next?' experience."

— Warren Keuffel, *Software Development Magazine*

"Beyond the engaging style that drags you forward from know-nothing into exalted Java warrior status, *Head First Java* covers a huge amount of practical matters that other texts leave as the dreaded 'exercise for the reader.…' It's clever, wry, hip and practical—there aren't a lot of textbooks that can make that claim and live up to it while also teaching you about object serialization and network launch protocols."

— Dr. Dan Russell, Director of User Sciences and Experience Research IBM Almaden Research Center (and teaches Artificial Intelligence at Stanford University)

"It's fast, irreverent, fun, and engaging. Be careful—you might actually learn something!"

— Ken Arnold, former Senior Engineer at Sun Microsystems Coauthor (with James Gosling, creator of Java), *The Java Programming Language*

"I feel like a thousand pounds of books have just been lifted off of my head."

— Ward Cunningham, inventor of the Wiki and founder of the Hillside Group

"Just the right tone for the geeked-out, casual-cool guru coder in all of us. The right reference for practical development strategies—gets my brain going without having to slog through a bunch of tired, stale professor-speak."

— Travis Kalanick, founder of Scour and Red Swoosh Member of the MIT TR100

"There are books you buy, books you keep, books you keep on your desk, and thanks to O'Reilly and the Head First crew, there is the penultimate category, Head First books. They're the ones that are dog-eared, mangled, and carried everywhere. *Head First SQL* is at the top of my stack. Heck, even the PDF I have for review is tattered and torn."

— Bill Sawyer, ATG Curriculum Manager, Oracle

"This book's admirable clarity, humor and substantial doses of clever make it the sort of book that helps even non-programmers think well about problem-solving."

— Cory Doctorow, co-editor of Boing Boing Author, *Down and Out in the Magic Kingdom* and *Someone Comes to Town, Someone Leaves Town*

Praise for other *Head First* books

"I received the book yesterday and started to read it…and I couldn't stop. This is definitely très 'cool.' It is fun, but they cover a lot of ground and they are right to the point. I'm really impressed."

— Erich Gamma, IBM Distinguished Engineer, and coauthor of *Design Patterns*

"One of the funniest and smartest books on software design I've ever read."

— Aaron LaBerge, VP Technology, ESPN.com

"What used to be a long trial and error learning process has now been reduced neatly into an engaging paperback."

— Mike Davidson, CEO, Newsvine, Inc.

"Elegant design is at the core of every chapter here, each concept conveyed with equal doses of pragmatism and wit."

— Ken Goldstein, Executive Vice President, Disney Online

"I ♥ *Head First HTML with CSS & XHTML*—it teaches you everything you need to learn in a 'fun-coated' format."

— Sally Applin, UI Designer and Artist

"Usually when reading through a book or article on design patterns, I'd have to occasionally stick myself in the eye with something just to make sure I was paying attention. Not with this book. Odd as it may sound, this book makes learning about design patterns fun.

"While other books on design patterns are saying 'Bueller…Bueller…Bueller…' this book is on the float belting out 'Shake it up, baby!'"

— Eric Wuehler

"I literally love this book. In fact, I kissed this book in front of my wife."

— Satish Kumar

Other related books from O'Reilly

Learning Python

Programming Python

Python in a Nutshell

Python Cookbook

Python for Unix and Linux System Administration

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Paul Barry

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Head First Python

by Paul Barry

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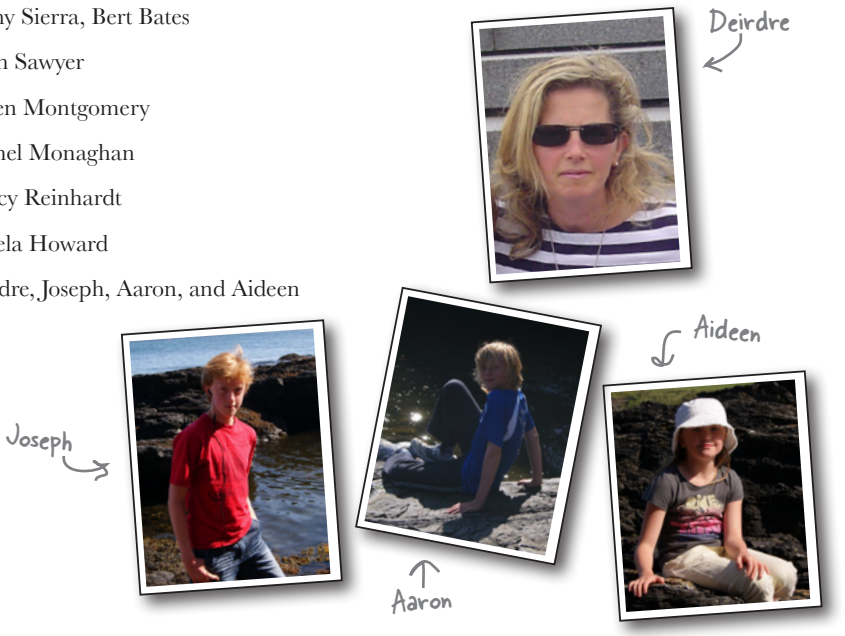

No athletes were pushed too hard in the making of this book.

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[M]

I dedicate this book to all those generous people in the Python community who have helped to make this great little language the *first-rate* programming technology it is.

And to those that made learning Python and its technologies just complex enough that people need a book like this to learn it.

Author of Head First Python

Paul Barry recently worked out that he has been programming for close to a quarter century, a fact that came as a bit of a shock. In that time, Paul has programmed in lots of different programming languages, lived and worked in two countries on two continents, got married, had three kids (well…his wife Deirdre actually *had them*, but Paul was there), completed a B.Sc. and M.Sc. in Computing, written or cowritten three other books, as well as a bunch of technical articles for *Linux Journal* (where he's a Contributing Editor).

When Paul first saw *Head First HTML with CSS & XHTML*, he loved it so much he knew immediately that the Head First approach would be a great way to teach programming. He was only too delighted then, together with David Griffiths, to create *Head First Programming* in an attempt to prove his hunch correct.

Paul's day job is working as a lecturer at The Institute of Technology, Carlow, in Ireland. As part of the Department of Computing and Networking, Paul gets to spend his day exploring, learning, and teaching programming technologies to his students, including Python.

Paul recently completed a post-graduate certificate in Learning and Teaching and was more than a bit relieved to discover that most of what he does conforms to current third-level best practice.

Table of Contents (Summary)

Table of Contents (the real thing)

Intro

Your brain on Python. Here you are trying to learn something, while here your brain is doing you a favor by making sure the learning doesn't stick. Your brain's thinking, "Better leave room for more important things, like which wild animals to avoid and whether naked snowboarding is a bad idea." So how do you trick your brain into thinking that your life depends on knowing Python?

meet python

1 Everyone loves lists

You're asking one question: "What makes Python different?"

The short answer is: *lots of things*. The longer answers starts by stating that there's lots that's familiar, too. Python is a lot like any other *general-purpose* programming language, with **statements**, **expressions**, **operators**, **functions**, **modules**, **methods**, and **classes**. All the *usual stuff*, really. And then there's the other stuff Python provides that makes the programmer's life—y*our* life—that little bit easier. You'll start your tour of Python by learning about **lists**. But, before getting to that, there's another important question that needs answering…

The Holy Grail, 1975, Terry Jones & Terry Gilliam, 91 mins

Graham Chapman

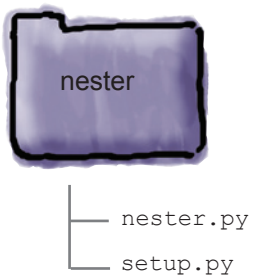
Michael Palin, John Cleese, Terry Gilliam, Eric Idle & Terry Jones

sharing your code

2 Modules of functions

Reusable code is great, but a shareable module is better.

By sharing your code as a Python module, you open up your code to the entire Python community…and it's always good to share, isn't it? In this chapter, you'll learn how to create, install, and distribute your own shareable modules. You'll then load your module onto Python's software sharing site on the Web, so that *everyone* can benefit from your work. Along the way, you'll pick up a few new tricks relating to Python's functions, too.

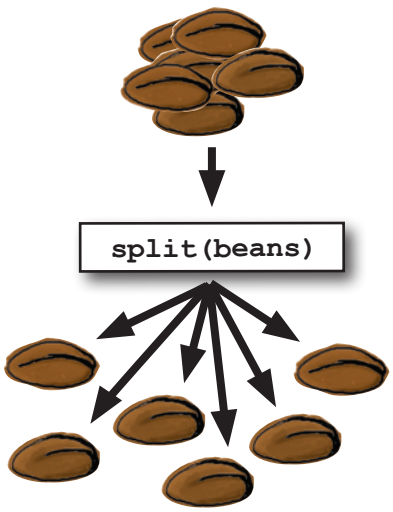


3 files and exceptions

Dealing with errors

It's simply not enough to process your list data in your code.

You need to be able to get your data *into* your programs with ease, too. It's no surprise then that Python makes reading data from **files** easy. Which is great, until you consider what can go *wrong* when interacting with data *external* to your programs… and there are lots of things waiting to trip you up! When bad stuff happens, you need a strategy for getting out of trouble, and one such strategy is to deal with any exceptional situations using Python's **exception handling** mechanism showcased in this chapter.

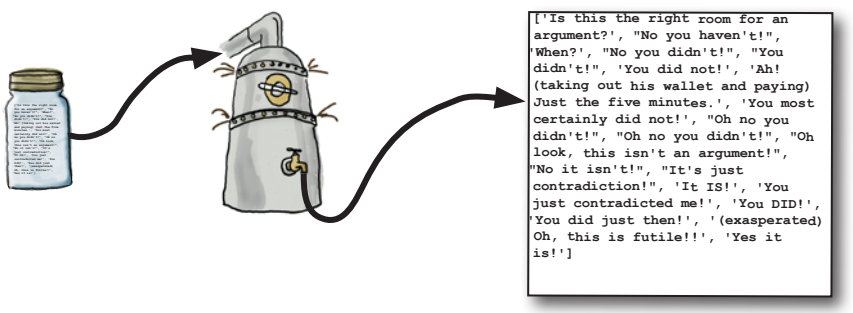


persistence

4 Saving data to files

It is truly great to be able to process your file-based data.

But what happens to your data when you're done? Of course, it's best to save your data to a disk file, which allows you to use it again at some later date and time. Taking your memory-based data and storing it to disk is what **persistence** is all about. Python supports all the usual tools for writing to files and also provides some cool facilities for *efficiently* storing Python data.



5

comprehending data

Work that data!

Data comes in all shapes and sizes, formats and encodings.

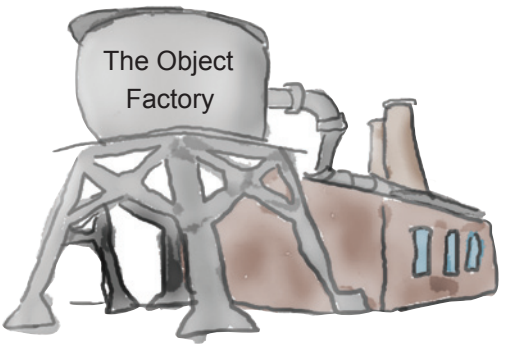
To work effectively with your data, you often have to manipulate and transform it into a common format to allow for efficient processing, sorting, and storage. In this chapter, you'll explore Python goodies that help you work your data up into a sweat, allowing you to achieve data-munging greatness.

custom data objects

6 Bundling code with data

It's important to match your data structure choice to your data.

And that choice can make a big difference to the complexity of your code. In Python, although really useful, lists and sets aren't the only game in town. The Python **dictionary** lets you organize your data for speedy lookup by *associating your data with names*, not numbers. And when Python's built-in data structures don't quite cut it, the Python **class** statement lets you define your own. This chapter shows you how.

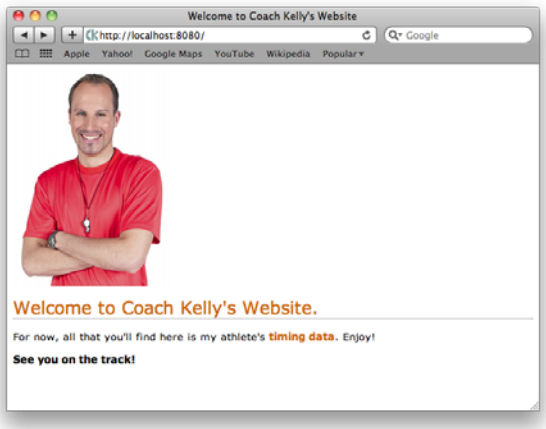


web development

7 Putting it all together

Sooner or later, you'll want to share your app with lots of people.

You have many options for doing this. Pop your code on PyPI, send out lots of emails, put your code on a CD or USB, or simply install your app manually on the computers of those people who need it. Sounds like a lot of work…not to mention boring. Also, what happens when you produce the next best version of your code? What happens then? How do you manage the update? Let's face it: it's such a pain that you'll think up really creative excuses not to. Luckily, you don't have to do any of this: just create a webapp instead. And, as this chapter demonstrates, using Python for web development is a breeze.



mobile app development

8 Small devices

Putting your data on the Web opens up all types of possibilities.

Not only can anyone from anywhere interact with your webapp, but they are increasingly doing so from a collection of diverse computing devices: PCs, laptops, tablets, palmtops, and even mobile phones. And it's not just humans interacting with your webapp that you have to support and worry about: *bots* are small programs that can automate web interactions and typically want your data, not your human-friendly HTML. In this chapter, you exploit Python on Coach Kelly's mobile phone to write an app that interacts with your webapp's data.

manage your data

9 Handling input

The Web and your phone are not just great ways to display data.

They are also great tools to for accepting input from your users. Of course, once your webapp accepts data, it needs to put it somewhere, and the choices you make when deciding what and where this "somewhere" is are often the difference between a webapp that's easy to grow and extend and one that isn't. In this chapter, you'll extend your webapp to accept data from the Web (via a browser or from an Android phone), as well as look at and enhance your back-end data-management services.

scaling your webapp

10 Getting real

The Web is a great place to host your app…until things get real.

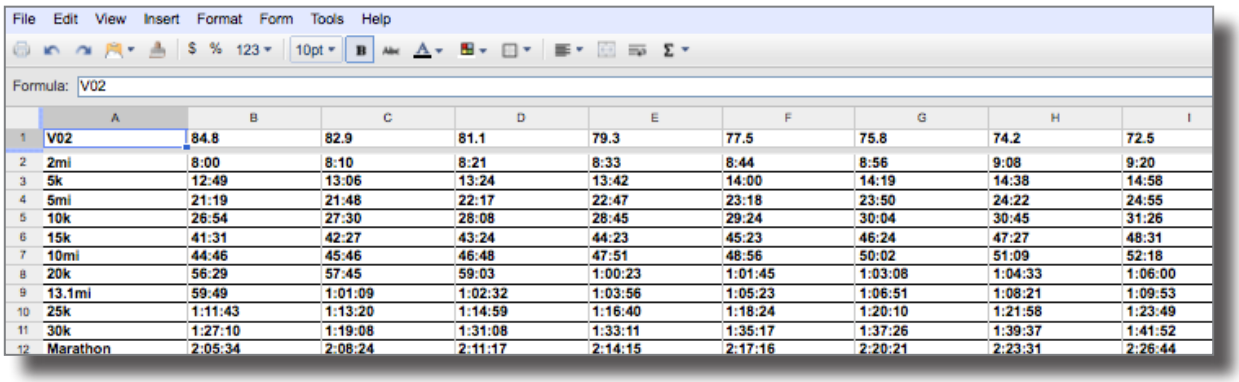
Sooner or later, you'll hit the jackpot and your webapp will be *wildly successful*. When that happens, your webapp goes from a handful of hits a day to thousands, possibly ten of thousands, *or even more*. Will you be ready? Will your web server handle the **load**? How will you know? What will it **cost**? *Who will pay?* Can your data model scale to millions upon millions of data items without *slowing to a crawl*? Getting a webapp up and running is easy with Python and now, thanks to Google App Engine, **scaling** a Python webapp is achievable, too.

dealing with complexity

11 Data wrangling

It's great when you can apply Python to a specific domain area.

Whether it's *web development*, *database management*, or *mobile apps*, Python helps you **get the job done** by *not* getting in the way of you coding your solution. And then there's the other types of problems: the ones you can't categorize or attach to a domain. Problems that are in themselves so *unique* you have to look at them in a different, highly specific way. Creating **bespoke** software solutions to these type of problems is an area where Python *excels*. In this, your final chapter, you'll *stretch* your Python skills to the limit and solve problems along the way.



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4	5mi	21:19	21:48	22:17	22:47	23:18	23:50	24:22	24:55
5	10k	26:54	27:30	28:08	28:45	29:24	30:04	30:45	31:26
6	15k	41:31	42:27	43:24	44:23	45:23	46:24	47:27	48:31
7	10mi	44:46	45:46	46:48	47:51	48:56	50:02	51:09	52:18
8	20k	56:29	57:45	59:03	1:00:23	1:01:45	1:03:08	1:04:33	1:06:00
9	13.1mi	59:49	1:01:09	1:02:32	1:03:56	1:05:23	1:06:51	1:08:21	1:09:53
10	25k	1:11:43	1:13:20	1:14:59	1:16:40	1:18:24	1:20:10	1:21:58	1:23:49
11	30k	1:27:10	1:19:08	1:31:08	1:33:11	1:35:17	1:37:26	1:39:37	1:41:52
12	Marathon	2:05:34	2:08:24	2:11:17	2:14:15	2:17:16	2:20:21	2:23:31	2:26:44

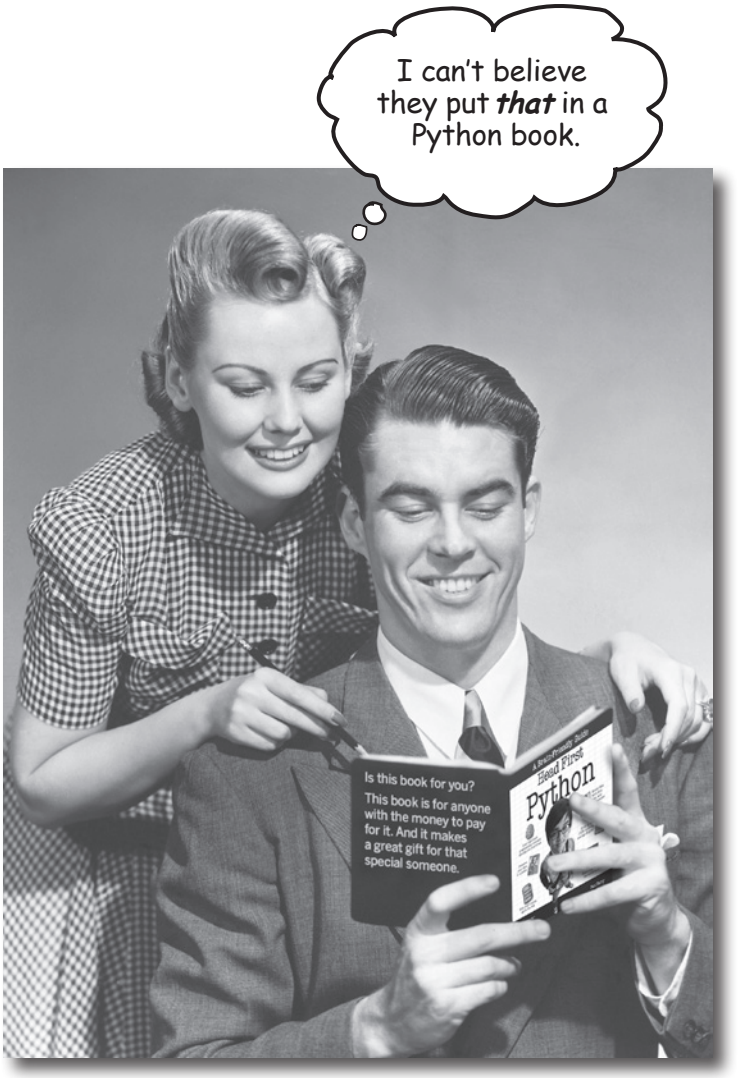
leftovers

The Top Ten Things (we didn't cover)

You've come a long way.

But learning about Python is an activity that never stops. The more Python you code, the more you'll need to learn new ways to do certain things. You'll need to master new tools and new techniques, too. There's just not enough room in this book to show you everything you might possibly need to know about Python. So, here's our list of the top ten things we didn't cover that you might want to learn more about next.

Intro



In this section, we answer the burning question:
"So why DID they put that in a Python book?"

Who is this book for?

If you can answer "yes" to all of these:

1. Do you already know how to program in another programming language?
2. Do you wish you had the know-how to program Python, add it to your list of tools, and make it do new things?
3. Do you prefer actually doing things and applying the stuff you learn over listening to someone in a lecture rattle on for hours on end?

this book is for you.

Who should probably back away from this book?

If you can answer "yes" to any of these:

1. Do you already know most of what you need to know to program with Python?
2. Are you looking for a reference book to Python, one that covers all the details in excruciating detail?
3. Would you rather have your toenails pulled out by 15 screaming monkeys than learn something new? Do you believe a Python book should cover *everything* and if it bores the reader to tears in the process then so much the better?

this book is ***not*** for you.

[Note from marketing: this book is for anyone with a credit card... we'll accept a check, too.]

We know what you're thinking

"How can *this* be a serious Python book?"

"What's with all the graphics?"

"Can I actually *learn* it this way?"

We know what your *brain* is thinking

Your brain craves novelty. It's always searching, scanning, *waiting* for something unusual. It was built that way, and it helps you stay alive.

So what does your brain do with all the routine, ordinary, normal things you encounter? Everything it *can* to stop them from interfering with the brain's *real* job—recording things that *matter*. It doesn't bother saving the boring things; they never make it past the "this is obviously not important" filter.

How does your brain *know* what's important? Suppose you're out for a day hike and a tiger jumps in front of you, what happens inside your head and body?

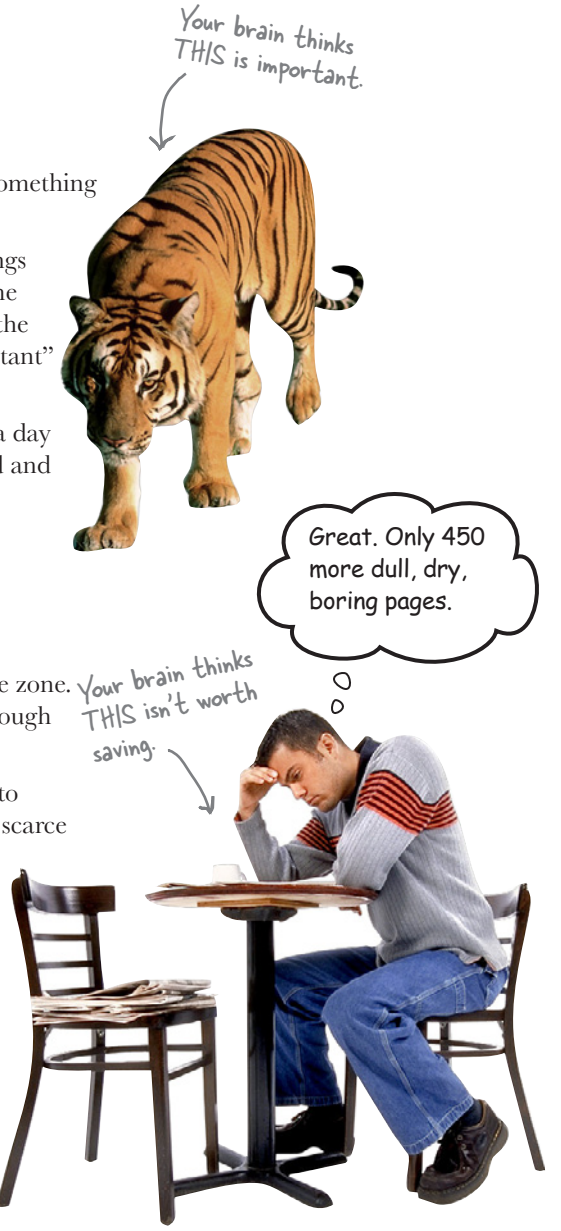
Neurons fire. Emotions crank up. *Chemicals surge*.

And that's how your brain knows…

This must be important! Don't forget it!

But imagine you're at home, or in a library. It's a safe, warm, tiger-free zone. You're studying. Getting ready for an exam. Or trying to learn some tough technical topic your boss thinks will take a week, ten days at the most.

Just one problem. Your brain's trying to do you a big favor. It's trying to make sure that this *obviously* non-important content doesn't clutter up scarce resources. Resources that are better spent storing the really *big* things. Like tigers. Like the danger of fire. Like how you should never have posted those "party" photos on your Facebook page. And there's no simple way to tell your brain, "Hey brain, thank you very much, but no matter how dull this book is, and how little I'm registering on the emotional Richter scale right now, I really *do* want you to keep this stuff around."



We think of a "Head First" reader as a learner.

So what does it take to *learn* something? First, you have to *get* it, then make sure you don't *forget* it. It's not about pushing facts into your head. Based on the latest research in cognitive science, neurobiology, and educational psychology, *learning* takes a lot more than text on a page. We know what turns your brain on.

Some of the Head First learning principles:

Make it visual. Images are far more memorable than words alone, and make learning much more effective (up to 89% improvement in recall and transfer studies). It also makes things more understandable. **Put the words within or near the graphics** they relate to, rather than on the bottom or on another page, and learners will be up to *twice* as likely to solve problems related to the content.

Use a conversational and personalized style. In recent studies, students performed up to 40% better on post-learning tests if the content spoke directly to the reader, using a first-person, conversational style rather than taking a formal tone. Tell stories instead of lecturing. Use casual language. Don't take yourself too seriously. Which would *you* pay more attention to: a stimulating dinner party companion or a lecture?

Get the learner to think more deeply. In other words, unless you actively flex your neurons, nothing much happens in your head. A reader has to be motivated, engaged, curious, and inspired to solve problems, draw conclusions, and generate new knowledge. And for that, you need challenges, exercises, and thought-provoking questions, and activities that involve both sides of the brain and multiple senses.

Get—and keep—the reader's attention. We've all had the "I really want to learn this but I can't stay awake past page one" experience. Your brain pays attention to things that are out of the ordinary, interesting, strange, eye-catching, unexpected. Learning a new, tough, technical topic doesn't have to be boring. Your brain will learn much more quickly if it's not.

Touch their emotions. We now know that your ability to remember something is largely dependent on its emotional content. You remember what you care about. You remember when you *feel* something. No, we're not talking heart-wrenching stories about a boy and his dog. We're talking emotions like surprise, curiosity, fun, "what the…?", and the feeling of "I Rule!" that comes when you solve a puzzle, learn something everybody else thinks is hard, or realize you know something that "I'm more technical than thou" Bob from engineering *doesn't*.

Metacognition: thinking about thinking

If you really want to learn, and you want to learn more quickly and more deeply, pay attention to how you pay attention. Think about how you think. Learn how you learn.

Most of us did not take courses on metacognition or learning theory when we were growing up. We were *expected* to learn, but rarely *taught* to learn.

But we assume that if you're holding this book, you really want to learn how to design user-friendly websites. And you probably don't want to spend a lot of time. If you want to use what you read in this book, you need to *remember* what you read. And for that, you've got to *understand* it. To get the most from this book, or *any* book or learning experience, take responsibility for your brain. Your brain on *this* content.

The trick is to get your brain to see the new material you're learning as Really Important. Crucial to your well-being. As important as a tiger. Otherwise, you're in for a constant battle, with your brain doing its best to keep the new content from sticking.

So just how *DO* you get your brain to treat programming like it was a hungry tiger?

There's the slow, tedious way, or the faster, more effective way. The slow way is about sheer repetition. You obviously know that you *are* able to learn and remember even the dullest of topics if you keep pounding the same thing into your brain. With enough repetition, your brain says, "This doesn't *feel* important to him, but he keeps looking at the same thing *over* and *over* and *over*, so I suppose it must be."

The faster way is to do ***anything that increases brain activity,*** especially different *types* of brain activity. The things on the previous page are a big part of the solution, and they're all things that have been proven to help your brain work in your favor. For example, studies show that putting words *within* the pictures they describe (as opposed to somewhere else in the page, like a caption or in the body text) causes your brain to try to makes sense of how the words and picture relate, and this causes more neurons to fire. More neurons firing = more chances for your brain to *get* that this is something worth paying attention to, and possibly recording.

A conversational style helps because people tend to pay more attention when they perceive that they're in a conversation, since they're expected to follow along and hold up their end. The amazing thing is, your brain doesn't necessarily *care* that the "conversation" is between you and a book! On the other hand, if the writing style is formal and dry, your brain perceives it the same way you experience being lectured to while sitting in a roomful of passive attendees. No need to stay awake.

But pictures and conversational style are just the beginning…

Here's what WE did:

We used ***pictures***, because your brain is tuned for visuals, not text. As far as your brain's concerned, a picture really *is* worth a thousand words. And when text and pictures work together, we embedded the text *in* the pictures because your brain works more effectively when the text is *within* the thing the text refers to, as opposed to in a caption or buried in the text somewhere.

We used ***redundancy***, saying the same thing in *different* ways and with different media types, and *multiple senses*, to increase the chance that the content gets coded into more than one area of your brain.

We used concepts and pictures in ***unexpected*** ways because your brain is tuned for novelty, and we used pictures and ideas with at least *some* ***emotional*** *content*, because your brain is tuned to pay attention to the biochemistry of emotions. That which causes you to *feel* something is more likely to be remembered, even if that feeling is nothing more than a little ***humor***, ***surprise***, or ***interest***.

We used a personalized, ***conversational style***, because your brain is tuned to pay more attention when it believes you're in a conversation than if it thinks you're passively listening to a presentation. Your brain does this even when you're *reading*.

We included more than 80 ***activities***, because your brain is tuned to learn and remember more when you ***do*** things than when you *read* about things. And we made the exercises challenging-yet-do-able, because that's what most people prefer.

We used ***multiple learning styles***, because *you* might prefer step-by-step procedures, while someone else wants to understand the big picture first, and someone else just wants to see an example. But regardless of your own learning preference, *everyone* benefits from seeing the same content represented in multiple ways.

We include content for ***both sides of your brain***, because the more of your brain you engage, the more likely you are to learn and remember, and the longer you can stay focused. Since working one side of the brain often means giving the other side a chance to rest, you can be more productive at learning for a longer period of time.

And we included ***stories*** and exercises that present ***more than one point of view,*** because your brain is tuned to learn more deeply when it's forced to make evaluations and judgments.

We included ***challenges***, with exercises, and by asking ***questions*** that don't always have a straight answer, because your brain is tuned to learn and remember when it has to *work* at something. Think about it—you can't get your *body* in shape just by *watching* people at the gym. But we did our best to make sure that when you're working hard, it's on the *right* things. That ***you're not spending one extra dendrite*** processing a hard-to-understand example, or parsing difficult, jargon-laden, or overly terse text.

We used ***people***. In stories, examples, pictures, etc., because, well, because *you're* a person. And your brain pays more attention to *people* than it does to *things*.

Cut this out and stick it on your refrigerator.

Here's what YOU can do to bend your brain into submission

So, we did our part. The rest is up to you. These tips are a starting point; listen to your brain and figure out what works for you and what doesn't. Try new things.

1. **Slow down. The more you understand, the less you have to memorize.**
Don't just *read*. Stop and think. When the book asks you a question, don't just skip to the answer. Imagine that someone really *is* asking the question. The more deeply you force your brain to think, the better chance you have of learning and remembering.

2. **Do the exercises. Write your own notes.**
We put them in, but if we did them for you, that would be like having someone else do your workouts for you. And don't just *look* at the exercises. **Use a pencil.** There's plenty of evidence that physical activity *while* learning can increase the learning.

3. **Read the "There are No Dumb Questions."**
That means all of them. They're not optional sidebars, ***they're part of the core content!*** Don't skip them.

4. **Make this the last thing you read before bed. Or at least the last challenging thing.**
Part of the learning (especially the transfer to long-term memory) happens *after* you put the book down. Your brain needs time on its own, to do more processing. If you put in something new during that processing time, some of what you just learned will be lost.

5. **Talk about it. Out loud.**
Speaking activates a different part of the brain. If you're trying to understand something, or increase your chance of remembering it later, say it out loud. Better still, try to explain it out loud to someone else. You'll learn more quickly, and you might uncover ideas you hadn't known were there when you were reading about it.

6. **Drink water. Lots of it.**
Your brain works best in a nice bath of fluid. Dehydration (which can happen before you ever feel thirsty) decreases cognitive function.

7. **Listen to your brain.**
Pay attention to whether your brain is getting overloaded. If you find yourself starting to skim the surface or forget what you just read, it's time for a break. Once you go past a certain point, you won't learn faster by trying to shove more in, and you might even hurt the process.

8. **Feel something.**
Your brain needs to know that this *matters*. Get involved with the stories. Make up your own captions for the photos. Groaning over a bad joke is *still* better than feeling nothing at all.

9. **Write a lot of code!**
There's only one way to learn to program: **writing a lot of code**. And that's what you're going to do throughout this book. Coding is a skill, and the only way to get good at it is to practice. We're going to give you a lot of practice: every chapter has exercises that pose a problem for you to solve. Don't just skip over them—a lot of the learning happens when you solve the exercises. We included a solution to each exercise—don't be afraid to **peek at the solution** if you get stuck! (It's easy to get snagged on something small.) But try to solve the problem before you look at the solution. And definitely get it working before you move on to the next part of the book.

Read Me

This is a learning experience, not a reference book. We deliberately stripped out everything that might get in the way of learning whatever it is we're working on at that point in the book. And the first time through, you need to begin at the beginning, because the book makes assumptions about what you've already seen and learned.

This book is designed to get you up to speed with Python as quickly as possible.

As you need to know stuff, we teach it. So you won't find long lists of technical material, no tables of Python's operators, not its operator precedence rules. We don't cover *everything*, but we've worked really hard to cover the essential material as well as we can, so that you can get Python into your brain *quickly* and have it stay there. The only assumption we make is that you already know how to program in some other programming language.

This book targets Python 3

We use Release 3 of the Python programming language in this book, and we cover how to get and install Python 3 in the first chapter. That said, we don't completely ignore Release 2, as you'll discover in Chapters 8 through 11. But trust us, by then you'll be so happy using Python, you won't notice that the technologies you're programming are running Python 2.

We put Python to work for you right away.

We get you doing useful stuff in Chapter 1 and build from there. There's no hanging around, because we want you to be *productive* with Python right away.

The activities are NOT optional.

The exercises and activities are not add-ons; they're part of the core content of the book. Some of them are to help with memory, some are for understanding, and some will help you apply what you've learned. ***Don't skip the exercises.***

The redundancy is intentional and important.

One distinct difference in a Head First book is that we want you to *really* get it. And we want you to finish the book remembering what you've learned. Most reference books don't have retention and recall as a goal, but this book is about *learning*, so you'll see some of the same concepts come up more than once.

The examples are as lean as possible.

Our readers tell us that it's frustrating to wade through 200 lines of an example looking for the two lines they need to understand. Most examples in this book are shown within the smallest possible context, so that the part you're trying to learn is clear and simple. Don't expect all of the examples to be robust, or even complete—they are written specifically for learning, and aren't always fully functional.

We've placed a lot of the code examples on the Web so you can copy and paste them as needed. You'll find them at two locations:

http://www.headfirstlabs.com/books/hfpython/

http://python.itcarlow.ie

The Brain Power exercises don't have answers.

For some of them, there is no right answer, and for others, part of the learning experience of the Brain Power activities is for you to decide if and when your answers are right. In some of the Brain Power exercises, you will find hints to point you in the right direction.

The technical review team

David Griffiths

Phil Hartley

Jeremy Jones

Technical Reviewers:

David Griffiths is the author of *Head First Rails* and the coauthor of *Head First Programming*. He began programming at age 12, when he saw a documentary on the work of Seymour Papert. At age 15, he wrote an implementation of Papert's computer language LOGO. After studying Pure Mathematics at University, he began writing code for computers and magazine articles for humans. He's worked as an agile coach, a developer, and a garage attendant, but not in that order. He can write code in over 10 languages and prose in just one, and when not writing, coding, or coaching, he spends much of his spare time traveling with his lovely wife—and fellow *Head First* author—Dawn.

Phil Hartley has a degree in Computer Science from Edinburgh, Scotland. Having spent more than 30 years in the IT industry with specific expertise in OOP, he is now teaching full time at the University of Advancing Technology in Tempe, AZ. In his spare time, Phil is a raving NFL fanatic

Jeremy Jones is coauthor of *Python for Unix and Linux System Administration*. He has been actively using Python since 2001. He has been a developer, system administrator, quality assurance engineer, and tech support analyst. They all have their rewards and challenges, but his most challenging and rewarding job has been husband and father.

Acknowledgments

My editor:

Brian Sawyer was *Head First Python*'s editor. When not editing books, Brian likes to run marathons in his spare time. This turns out to be the perfect training for working on another book with me (our second together). O'Reilly and Head First are lucky to have someone of Brian's caliber working to make this and other books the best they can be.

The O'Reilly team:

Karen Shaner provided administrative support and very capably coordinated the techical review process, responding quickly to my many queries and requests for help. There's also the back-room gang to thank—the O'Reilly Production Team—who guided this book through its final stages and turned my InDesign files into the beautiful thing you're holding in your hands right now (or maybe you're on an iPad, Android tablet, or reading on your PC—that's cool, too).

And thanks to the other Head First authors who, via Twitter, offered cheers, suggestions, and encouragement throughout the entire writing process. You might not think 140 characters make a big difference, but they really do.

I am also grateful to **Bert Bates** who, together with **Kathy Sierra**, created this series of books with their wonderful *Head First Java*. At the start of this book, **Bert** took the time to set the tone with a marathon 90-minute phone call, which stretched my thinking on what I wanted to do to the limit and pushed me to write a better book. Now, some nine months after the phone call, I'm pretty sure I've recovered from the mind-bending **Bert** put me through.

Friends and colleagues:

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My students (those enrolled on 3rd Year Games Development and 4th Year Software Engineering) have been exposed to this material in various forms over the last 18 months. Their positive reaction to Python and the approach I take with my classes helped inform the structure and eventual content of this book. (And yes, folks, some of this is on your final).

Family:

My family, **Deirdre**, **Joseph**, **Aaron,** and **Aideen** had to, once more, bear the grunts and groans, huffs and puffs, and more than a few roars on more than one occasion (although, to be honest, not as often they did with *Head First Programming*). After the last book, I promised I wouldn't start another one "for a while." It turned out "a while" was no more than a few weeks, and I'll be forever grateful that they didn't gang up and throw me out of the house for breaking my promise. Without their support, and especially the ongoing love and support of my wife, Deirdre, this book would not have seen the light of day.

The without-whom list:

My technical review team did an excellent job of keeping me straight and making sure what I covered was spot on. They confirmed when my material was working, challenged me when it wasn't and not only pointed out when stuff was wrong, but provided suggestions on how to fix it. This is especially true of **David Griffiths**, my co-conspirator on *Head First Programming*, whose technical review comments went above and beyond the call of duty. **David's** name might not be on the cover of this book, but a lot of his ideas and suggestions grace its pages, and I was thrilled and will forever remain grateful that he approached his role as tech reviewer on *Head First Python* with such gusto.

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